

Primera práctica de Arquitectura en la Nube



***Extra de Docker al final**

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Utilizamos WSL (Windows Subsystem for Linux) para simular un entorno de servidor real basado en Linux. Esto es necesario porque la mayoría de los servidores en la nube operan con sistemas operativos basados en Linux

Instalación de WSL:

Instalación de WSL en PowerShell, con comando **wsl --install**, se instala de manera predeterminada Ubuntu y una vez completa la instalación te pedirá que reinicies el equipo. Al iniciar de vuelta y poner **wsl** en la línea de comando te pedirá que crees tu cuenta y pongas su respectiva contraseña.

```
PS C:\WINDOWS\system32> wsl --install
Descargando: Ubuntu
Instalando: Ubuntu
Distribución instalada correctamente. Se puede iniciar a través de "wsl.exe -d Ubuntu"
Iniciando Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Create a default Unix user account: rodrigo
New password:
Retype new password:
passwd: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$
```

Realizada la instalación accedemos y empezamos con todos los pasos del proyecto:

Paso 1: Actualización de los paquetes del sistema

```
PS C:\WINDOWS\system32> wsl
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt update
[sudo] password for rodrigo:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  liblvm2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  liblvm2
The following packages will be upgraded:
  coreutils dconf-gsettings-backend dconf-service landscape-client landscape-common libdconf1 libgl-mesa0 libglvnd
  libglx-mesa0 libglx-mesa0 libltd0 mesa-libgallium mesa-vulkan-drivers openssh-client
  python3-software-properties software-properties-common systemd-hue-hudb
17 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 59.2 MB/59.4 MB of archives.
After this operation, 144 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 coreutils amd64 9.4-3ubuntu0.1 [1413 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-hue-hudb all 235.1.5 [3518 B]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-client amd64 1:9.9p1-3ubuntu13.14 [906 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dconf-gsettings-backend amd64 0.40.0-4ubuntu0.1 [22.1 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dconf-service amd64 0.40.0-4ubuntu0.1 [27.6 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdconf1 amd64 0.40.0-4ubuntu0.1 [39.6 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 landscape-client amd64 24.02-0ubuntu0.6 [124 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 landscape-common amd64 24.02-0ubuntu0.6 [93.2 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libglx-mesa-dri amd64 25.0.7-0ubuntu0.24.04.2 [35.0 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libglx-mesa0 amd64 25.0.7-0ubuntu0.24.04.2 [141 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 liblvm2 amd64 1:2.03.12-0ubuntu0.24.04.2 [59.6 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgl-mesa0 amd64 25.0.7-0ubuntu0.24.04.2 [124 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libglvnd amd64 25.0.7-0ubuntu0.24.04.2 [32.7 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 mesa-libgallium amd64 25.0.7-0ubuntu0.24.04.2 [10.3 MB]
Get:15 [4 mesa-libgallium 16.4 kB/10.3 MB OK]_
```

[illegible]

```

rodriago@A6Alumno11: /mnt/c/WINDOWS/system32$ sudo service apache2 start
rodriago@A6Alumno11: /mnt/c/WINDOWS/system32$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-09-30 09:28:59 CEST; 1min 56s ago
     Docs: https://httpd.apache.org/docs/2.4/
    Process: 10492 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 10495 (apache2)
      Tasks: 6 (limit: 9350)
     Memory: 10.5M (peak: 12.4M)
        CPU: 38ms
    CGroup: /system.slice/apache2.service
            └─10495 /usr/sbin/apache2 -k start
              └─10497 /usr/sbin/apache2 -k start
                └─10499 /usr/sbin/apache2 -k start
                  └─10500 /usr/sbin/apache2 -k start
                    └─10501 /usr/sbin/apache2 -k start
                      └─10502 /usr/sbin/apache2 -k start

Sep 30 09:28:59 A6Alumno11 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 30 09:28:59 A6Alumno11 systemd[1]: Started apache2.service - The Apache HTTP Server.
rodriago@A6Alumno11: /mnt/c/WINDOWS/system32$

```

Paso 4: Creación HTML simple y pruebas del HTML desde terminal y en navegador

```
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ echo "<?php phpinfo(); ?>" | sudo tee /var/www/html/info.php
<?php phpinfo(); ?>
```

[illegible]

PHP Version 8.3.6	
	
System	Linux A6Aumt11 6.8.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64
Build Date	Jul 14 2025 18:30:55
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.3/apache2
Loaded Configuration File	/etc/php/8.3/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.3/apache2/conf.d
Additional .ini files parsed	/etc/php/8.3/apache2/conf.d/10-opcache.ini, /etc/php/8.3/apache2/conf.d/10-pdo.ini, /etc/php/8.3/apache2/conf.d/20-calendar.ini, /etc/php/8.3/apache2/conf.d/20-ctype.ini, /etc/php/8.3/apache2/conf.d/20-enlme.ini, /etc/php/8.3/apache2/conf.d/20-fn.ini, /etc/php/8.3/apache2/conf.d/20-fileinfo.ini, /etc/php/8.3/apache2/conf.d/20-gdlib.ini, /etc/php/8.3/apache2/conf.d/20-gettext.ini, /etc/php/8.3/apache2/conf.d/20-iconv.ini, /etc/php/8.3/apache2/conf.d/20-phar.ini, /etc/php/8.3/apache2/conf.d/20-posix.ini, /etc/php/8.3/apache2/conf.d/20-readline.ini, /etc/php/8.3/apache2/conf.d/20-shmop.ini, /etc/php/8.3/apache2/conf.d/20-sysctl.ini, /etc/php/8.3/apache2/conf.d/20-sysmsg.ini, /etc/php/8.3/apache2/conf.d/20-sysvshm.ini, /etc/php/8.3/apache2/conf.d/20-tokenizer.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	20230831
Zend Extension Build	APM20230831.NTS
PHP Extension Build	APM20230831.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	disabled
Zend Max Execution Times	disabled
IPv6 Support	enabled
DTrace Support	disabled
Registered PHP Streams	https, ftps, compress.zlib, php, file, glob, data, http, ftp, phar
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, tls, IPv6, l, bev, i, stream, r, tcp, udp6
Registered Stream Filters	zlib.*, string.rot13, string.toupper, string.tolower, convert.*, consumed, dechunk, convert.iconv.*
This program makes use of the Zend Scripting Language Engine: Zend engine v3.6.6, Copyright (c) Zend Technologies with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies 	

Detenemos el servicio Apache. Es fundamental liberar el puerto 80, ya que dos servicios (Apache y Nginx) no pueden escucharlo simultáneamente. Esto asegura que Nginx pueda iniciarse sin conflictos.

Usando **'service apache2 stop'** podemos ver que el servicio queda inactivo

```
root@A6Alumno11:/mnt/c/WINDOWS/system32# service apache2 stop
```

```
root@A6Alumno11:/mnt/c/WINDOWS/system32# service apache2 stop
root@A6Alumno11:/mnt/c/WINDOWS/system32# systemctl status apache2
○ apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: inactive (dead) since Tue 2025-09-30 14:46:28 CEST; 23s ago
     Duration: 17.633s
    Docs: https://httpd.apache.org/docs/2.4/
   Process: 9896 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Process: 9941 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, status=0/SUCCESS)
   Main PID: 9899 (code=exited, status=0/SUCCESS)
      CPU: 60ms

Sep 30 14:46:10 A6Alumno11 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Sep 30 14:46:10 A6Alumno11 systemd[1]: Started apache2.service - The Apache HTTP Server.
Sep 30 14:46:28 A6Alumno11 systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Sep 30 14:46:28 A6Alumno11 systemd[1]: apache2.service: Deactivated successfully.
Sep 30 14:46:28 A6Alumno11 systemd[1]: Stopped apache2.service - The Apache HTTP Server.
root@A6Alumno11:/mnt/c/WINDOWS/system32#
```

Paso 1: Instalamos el servidor web Nginx

```
root@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libltdl0
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 564 kB in 0s (1848 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 41790 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Could not execute systemctl: at /usr/bin/deb-systemd-invoke line 148.
Setting up nginx (1.24.0-2ubuntu7.5) ...
Not attempting to start NGINX, port 80 is already in use.
Processing triggers for man-db (2.12.0-4build2) ...
```

Paso 2: Una vez instalado, se inicia el servicio de la misma manera que hicimos con apache.

También verificamos que el servicio está corriendo sin problemas con **'systemctl status nginx'**.

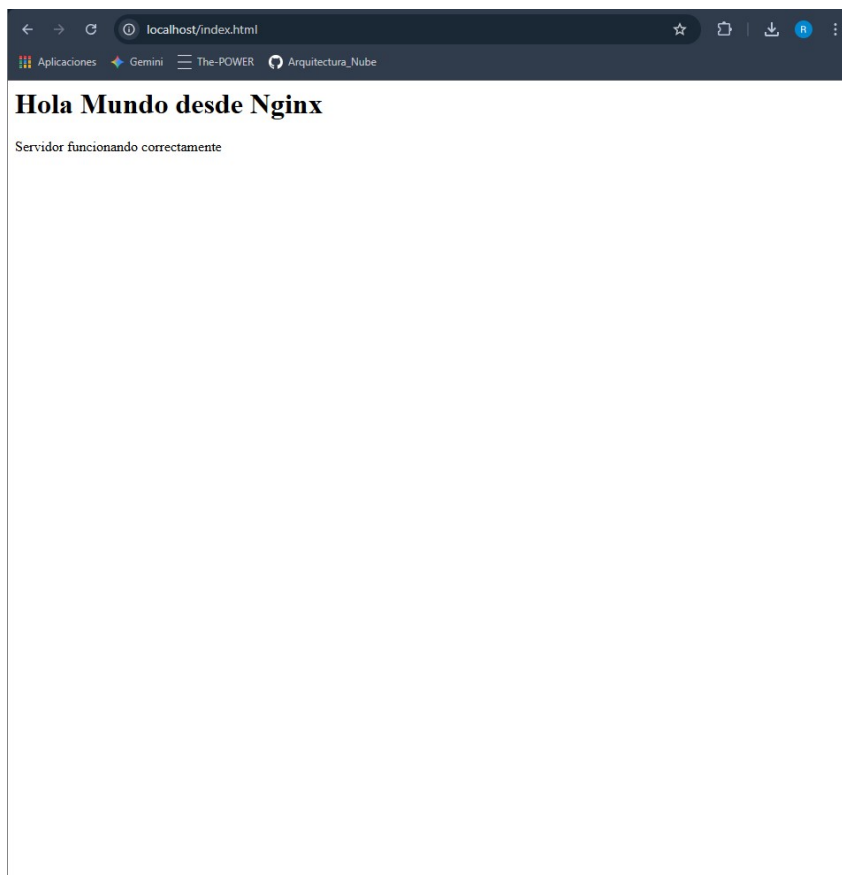
```
root@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo su
[sudo] password for rodrigo:
root@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo service nginx start
root@A6Alumno11:/mnt/c/WINDOWS/system32$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-09-30 10:12:21 CEST; 2min 19s ago
     Docs: man:nginx(8)
   Process: 175 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 207 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 216 (nginx)
      Tasks: 17 (limit: 9350)
     Memory: 12.8M (peak: 14.1M)
        CPU: 0ms
    CGroup: /system.slice/nginx.service
           └─216 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─217 "nginx: worker process"
               └─218 "nginx: worker process"
                 └─220 "nginx: worker process"
                   └─221 "nginx: worker process"
                     └─222 "nginx: worker process"
                       └─223 "nginx: worker process"
                         └─224 "nginx: worker process"
                           └─225 "nginx: worker process"
                             └─226 "nginx: worker process"
                               └─228 "nginx: worker process"
                                 └─229 "nginx: worker process"
                                   └─231 "nginx: worker process"
                                     └─234 "nginx: worker process"
                                       └─235 "nginx: worker process"
                                         └─236 "nginx: worker process"
                                           └─237 "nginx: worker process"

Sep 30 10:12:21 A6Alumno11 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Sep 30 10:12:21 A6Alumno11 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
lines 1-31/31 (END)
```

Paso 3: Creo una página HTML simple para poder hacer las comprobaciones requeridas

```
root@A6Alumno11:/mnt/c/WINDOWS/system32# echo "<h1>Hola Mundo desde Nginx</h1><p>Servidor funcionando correctamente</p>" | sudo tee /var/www/html/index.html
root@A6Alumno11:/mnt/c/WINDOWS/system32#
```

```
root@A6Alumno11:/mnt/c/WINDOWS/system32# ip addr show eth0 | grep inet
    inet 172.24.98.57/20 brd 172.24.111.255 scope global eth0
    inet6 fe80::215:5dff:fea3:9d31/64 scope link
root@A6Alumno11:/mnt/c/WINDOWS/system32#
```



PRACTICA EXTRA CON DOCKER

Instalamos Docker para habilitar la ‘contenerización’. Docker permite empaquetar una aplicación con todas sus dependencias en un entorno aislado (el contenedor), garantizando que se ejecute de la misma manera en cualquier sistema

```

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt update && sudo apt upgrade -y
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Err:1 https://download.docker.com/linux/ubuntu noble InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7EA0A9C3F273FCD8
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
W: An error occurred during the signature verification. The repository is not updated and the previous index files will be used.
  GPG error: https://download.docker.com/linux/ubuntu noble InRelease: The following signatures couldn't be verified because the
public key is not available: NO_PUBKEY 7EA0A9C3F273FCD8
W: Failed to fetch https://download.docker.com/linux/ubuntu/dists/noble/InRelease The following signatures couldn't be verified
because the public key is not available: NO_PUBKEY 7EA0A9C3F273FCD8
W: Some index files failed to download. They have been ignored, or old ones used instead.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt install apt-transport-https ca-certificates curl software-properties-common
lsb-release -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apt-transport-https is already the newest version (2.8.3).
ca-certificates is already the newest version (20240203).
curl is already the newest version (8.5.0-2ubuntu10.6).
software-properties-common is already the newest version (0.99.49.3).
lsb-release is already the newest version (12.0-2).
lsb-release set to manually installed.
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQINBFit2ioBEADhWpZ8/vvZ6hUTiXOWQHXMAlaFHcPH9hAtr4F1y2+OYdbtMuth
lqqwp028AqyY+PRfVMtSYMbjuQuu5bbyKR01BbqYhuS3jtqQmljZ/b3vXqnm1VXh
38UuLa+z077PxyxQhu5BbqntTPQMfiyqEiU+8Kbq2WmANUKQf+1AmZY/IruOXbnq
L4C1+gJ8vfmXQt99npCaxEjaNRVYFOS8QcixNzHUYVnb6emjLANyEV1Zzeqo7XK17
UrwV5inawTSzWnvtjEjj4nJL8NsLwscLPQUhTQ+7BbQXAwAmeHCUTQIvvhXqW0N
cmhh4HgeQscQHYgO3jjDVfoY5MucvlgIgcqfzAHw9jxmRL4qbMzj+b1XoePEtHt
ku4bIQN1X5P07fNWzlgARL5Z4POXDDZTlIQ/EL58j9kp4bnWRCJW01ya+f8ocodo
vZZ+Doi+fy4D5Z6rL4XECIQP/Lv5uFyF+kQtL/94VFYVJ01eAv8W92KdgdKhtCTD
G7c0tIkVekNUq48b3aQ64NOZQW7fVjfoKwEzdOqPE72Pa45jrcZzvUFxSpdInk2tZ
XYukHj1xxEgBdC/J3cMMNRE1F4NCA3ApfV1Y7/hTeOnmDuDYwr9/obA8t016V1jj
q5rdkyuPF4JF8mXUW5eCN1vAFHxeg9ZMemhBtQmGxXnw9M+z6hWwc6ahmwARAQAB
tCtEb2NrXiGUmV5ZWZfZSZAoQ0UgZGVikaSA8ZG9ja2VyQGRvY2t1ci5jb20+1QI3
BBMBGcAhBQYyrefAAhsvBQsJCAcDBRUKCQGLBRYCAwEAAh4BAheAAAOJEI2BgDwO
vB2IsSkP/iQZo68fLDQmNvn8XSTd6RRaUH33kXYXquT6NKH3c1S7E2gTjmqqMqd
t4mMYHCSEYxI5qrcYV5Yq9P6+Ko+vozo4nseUQLPH/ATQ4qL0Zok+1jkag3Lgk
jonyUf9bwtWxPp05HC3GMHPPhcUSexCxQLQvnFWXD2sWLKivHp2fT8QbRGeZ+d3m
6Fqcd5Fu7pxsqm0EUDK5NL+nPiGyHn+auTrhgzhK1CSHfGccM/wfR1e19Utz6p9P
XRK1lWnXt4tqNGZNTN0tr+NLG/6Bqd80YBaFAUcue/w1VW6JQ2VGyZhnZuS8LMc

```


Para la instalación de docker, es recomendable seguir los pasos de la página web:

https://hub.docker.com/_/ubuntu.

Quizás algunos pasos estén desactualizados o den fallos y tuve que ayudarme de la inteligencia artificial para asegurar que la instalación fuera correcta.

```

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" > /etc/apt/sources.list.d/docker.list
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt update
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Get:2 https://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Err:3 https://download.docker.com/linux/ubuntu noble InRelease
  The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7EAB9C3F273FCD8
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [278 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [278 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [640 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7168 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 1025 kB in 1s (857 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
W: An error occurred during the signature verification. The repository is not updated and the previous index files will be used
GPG error: https://download.docker.com/linux/ubuntu noble InRelease: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7EAB9C3F273FCD8
W: Failed to fetch https://download.docker.com/linux/ubuntu/dists/noble/InRelease The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7EAB9C3F273FCD8
W: Some index files failed to download. They have been ignored, or old ones used instead.
rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker-ce is already the newest version (5:28.5.0-1-ubuntu.24.04-noble).
docker-ce-cli is already the newest version (5:28.5.0-1-ubuntu.24.04-noble).
containerd.io is already the newest version (1.7.28-0-ubuntu.24.04-noble).
docker-buildx-plugin is already the newest version (0.29.1-1-ubuntu.24.04-noble).
docker-compose-plugin is already the newest version (2.40.0-1-ubuntu.24.04-noble).
The following package was automatically installed and is no longer required:
  liblvm29
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.

```

```

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ docker pull ubuntu:latest
latest: Pulling from library/ubuntu
Digest: sha256:728785b5923d75e3e5c5af178fab1be7031f3522c5cd7a0b32b80d8248123
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest

```

Una vez instalado, verificamos su funcionamiento con systemctl

```

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ sudo systemctl status docker
docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-10-07 12:35:19 CEST; 14min ago
     Triggers: + docker.socket
       Docs: https://docs.docker.com
    Main PID: 363 (dockerd)
      Tasks: 16
     Memory: 81.6M (peak: 115.4M)
        CPU: 1.109s
     CGroup: /system.slice/docker.service
             └─363 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 07 12:35:19 A6Alumno11 dockerd[363]: time="2025-10-07T12:35:19.083960091+02:00" level=info msg="Loading containerd"
Oct 07 12:35:19 A6Alumno11 dockerd[363]: time="2025-10-07T12:35:19.097153476+02:00" level=info msg="Docker daemon" c
Oct 07 12:35:19 A6Alumno11 dockerd[363]: time="2025-10-07T12:35:19.097246516+02:00" level=info msg="Initializing buld
Oct 07 12:35:19 A6Alumno11 dockerd[363]: time="2025-10-07T12:35:19.118011788+02:00" level=info msg="Completed buildi
Oct 07 12:35:19 A6Alumno11 dockerd[363]: time="2025-10-07T12:35:19.128420206+02:00" level=info msg="Daemon has compl
Oct 07 12:35:19 A6Alumno11 dockerd[363]: time="2025-10-07T12:35:19.128473589+02:00" level=info msg="API listen on /r
Oct 07 12:35:19 A6Alumno11 systemd[1]: Started docker.service - Docker Application Container Engine.
Oct 07 12:39:45 A6Alumno11 dockerd[363]: time="2025-10-07T12:39:45.861830439+02:00" level=info msg="Ignoring event"
Oct 07 12:42:09 A6Alumno11 dockerd[363]: time="2025-10-07T12:42:09.566472333+02:00" level=info msg="Ignoring event"
Oct 07 12:49:41 A6Alumno11 dockerd[363]: time="2025-10-07T12:49:41.719120256+02:00" level=info msg="Ignoring event"
lines 1-22/22 (END)

```

Aquí creamos el contenedor con la última versión de ubuntu e iniciamos el contenedor para después entrar en su interfaz.

Una vez estemos dentro haremos los mismos pasos que en la anterior ocasión desde WSL y empezaremos por la instalación de apache2.

```

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ docker run -it --name final-server -p 8080:80 ubuntu:latest /bin/bash
root@207950dd30a0:/#

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS          PORTS          NAMES
207950dd30a0   ubuntu:latest  "/bin/bash"             9 minutes ago   Exited (0) 6 minutes ago          final-server

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ docker start 207950dd30a0
207950dd30a0

rodrigo@A6Alumno11:/mnt/c/WINDOWS/system32$ docker exec -it 207950dd30a0 /bin/bash
root@207950dd30a0:/#

```



```
root@20795dd30a0:/# apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2478 KB]
Get:3 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.6 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1519 KB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1138 KB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 KB]
Get:11 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1898 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1925 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2612 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1882 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [38.9 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [33.9 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.4 kB]
Fetched 33.9 MB in 3s (12.3 MB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
1 package can be upgraded. Run 'apt list --upgradable' to see it.
```

```
root@20795dd30a0:/# apt install apache2 php libapache2-mod-php -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  adduser apache2-bin apache2-data apache2-utils ca-certificates krb5-locales libapache2-mod-php8.3 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1-tls libargon2-1 libbrotli1 libbsd0 libcurl4t64 libedit2
  libexpat1 libgdbm-compat4t64 libgdbm6t64 libgssapi-krb5-2 libicu74 libjansson4 libk5crypto3 libkeyutils1
  libkrb5-3 libkrb5support0 libldap-common libldap2 liblua5.4-0 libnghttp2-14 libperl5.38t64 libpsl5t64 librtmp1
  libssl1.2 libssl2-modules libssl3-modules libsystemd0 libsystemd-shared libsystemd-sysv libsystemd-timesyncd libunistring2
  netbase openssl perl perl-modules-5.38 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline
  psmisc publicsuffix ssl-cert tzdata ucf
Suggested packages:
  liblocale-gettext-perl cron quota cryptfs-utils apache2-doc apache2-suexec-pristine | apache2-suexec-custom
  www-browser vdir-guard gdcm11en krb5-doc krb5-user libssl1.2-modules-gssapi-mit
  | libssl1.2-modules-gssapi-helios libssl1.2-modules-ldap libssl1.2-modules-otp libssl1.2-modules-sql perl-doc
  libterm-readline-gnu-perl | libterm-readline-perl-perl make libtap-harness-archive-perl
The following NEW packages will be installed:
  adduser apache2 apache2-bin apache2-data apache2-utils ca-certificates krb5-locales libapache2-mod-php
  libapache2-mod-php8.3 libaprutil1 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1-tls libargon2-1 libbrotli1
  libbsd0 libcurl4t64 libedit2 libexpat1 libgdbm-compat4t64 libgdbm6t64 libgssapi-krb5-2 libicu74 libjansson4
  libk5crypto3 libkeyutils1 libkrb5-3 libkrb5support0 libldap-common libldap2 liblua5.4-0 libnghttp2-14
  libperl5.38t64 libpsl5t64 librtmp1 libssl1.2 libssl2-modules libssl3-modules libsystemd0 libsystemd-shared
  libsystemd-sysv libsystemd-timesyncd libunistring2 netbase openssl perl perl-modules-5.38 php-common php8.3
  php8.3-cli php8.3-common php8.3-opcache php8.3-readline psmisc publicsuffix ssl-cert tzdata ucf
The following packages will be upgraded:
  libssl1.2t64
1 upgraded, 59 newly installed, 0 to remove and 0 not upgraded.
Need to get 34.6 MB of archives.
After this operation, 135 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-modules-5.38 all 5.38.2-3.ubuntu0.2 [3118 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libgdbm6t64 amd64 1.23.5-1build1 [34.4 kB]
```

```
root@20795dd30a0:/# service apache2 start
* Starting Apache httpd web server: apache2
AH00558: apache: could not reliably determine the server's fully qualified domain name, using 172.17.0.2. Set the 'ServerName' directive globally to suppress this message
*
root@20795dd30a0:/# service apache2 status
* apache2 is running
root@20795dd30a0:/# echo "<?php phpinfo(); ?>" > /var/www/html/info.php
root@20795dd30a0:/#
```

```
root@20795dd30a0:/# service apache2 status
* apache2 is running
```

[illegible]

NGINX EN DOCKER

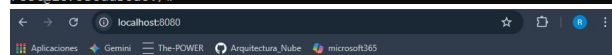
Apagamos el servicio de apache2 para liberar el puerto 80.

```
root@207950dd30a0:/# service apache2 stop
* Stopping Apache httpd web server apache2
*
root@207950dd30a0:/#
```

```
root@207950dd30a0:/# apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  iproute2 libatm1t64 libbpf1 libcap2-bin libelf1t64 libmnl0 libpam-cap libxtables12 nginx-common
Suggested packages:
  iproute2-doc python3:any fcgiwrap nginx-doc
The following NEW packages will be installed:
  iproute2 libatm1t64 libbpf1 libcap2-bin libelf1t64 libmnl0 libpam-cap libxtables12 nginx nginx-common
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 2025 kB of archives.
After this operation, 5799 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libelf1t64 amd64 0.190-1.1ubuntu0.1 [57.8 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libbpf1 amd64 1:1.3.0-2build2 [166 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libmnl0 amd64 1.0.5-2build1 [12.3 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libxtables12 amd64 1.8.10-3ubuntu2 [35.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcap2-bin amd64 1:2.66-5ubuntu2.2 [34.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 iproute2 amd64 6.1.0-1ubuntu6.2 [1120 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 libatm1t64 amd64 1:2.5.1-5.1build1 [22.9 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-cap amd64 1:2.66-5ubuntu2.2 [12.5 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]
Fetched 2025 kB in 0s (6246 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libelf1t64:amd64.
(Reading database ... 8673 files and directories currently installed.)
Preparing to unpack .../0-libelf1t64_0.190-1.1ubuntu0.1_amd64.deb ...
Unpacking libelf1t64:amd64 (0.190-1.1ubuntu0.1) ...
Selecting previously unselected package libbpf1:amd64.
Preparing to unpack .../1-libbpf1_1.3.0-2build2_amd64.deb ...
Unpacking libbpf1:amd64 (1:1.3.0-2build2) ...
Selecting previously unselected package libmnl0:amd64.
Preparing to unpack .../2-libmnl0_1.0.5-2build1_amd64.deb ...
Unpacking libmnl0:amd64 (1.0.5-2build1) ...
Selecting previously unselected package libxtables12:amd64.
Preparing to unpack .../3-libxtables12_1.8.10-3ubuntu2_amd64.deb ...
Unpacking libxtables12:amd64 (1.8.10-3ubuntu2) ...
Selecting previously unselected package libcap2-bin.
Preparing to unpack .../4-libcap2-bin_1:2.66-5ubuntu2.2_amd64.deb ...
Unpacking libcap2-bin (1:2.66-5ubuntu2.2) ...
Selecting previously unselected package iproute2.
Preparing to unpack .../6-iproute2_6.1.0-1ubuntu6.2_amd64.deb ...
Unpacking iproute2 (6.1.0-1ubuntu6.2) ...
Selecting previously unselected package libatm1t64:amd64.
Preparing to unpack .../6-libatm1t64_1:2.5.1-5.1build1_amd64.deb ...
Unpacking libatm1t64:amd64 (1:2.5.1-5.1build1) ...
Selecting previously unselected package libpam-cap:amd64.
Preparing to unpack .../7-libpam-cap_1:2.66-5ubuntu2.2_amd64.deb ...
Unpacking libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
Selecting previously unselected package nginx-common.
Preparing to unpack .../8-nginx-common_1.24.0-2ubuntu7.5_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.5) ...
Selecting previously unselected package nginx.
Preparing to unpack .../9-nginx_1.24.0-2ubuntu7.5_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.5) ...
Setting up libatm1t64:amd64 (1:2.5.1-5.1build1) ...
Setting up nginx-common (1.24.0-2ubuntu7.5) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl
```

```
root@207950dd30a0:/# service nginx start
* Starting nginx nginx
root@207950dd30a0:/# service nginx status
* nginx is running
root@207950dd30a0:/#
```

```
root@207950dd30a0:/# echo "<h1>Hola Mundo desde Nginx en Docker</h1><p>Servidor funcionando correctamente</p>" > /var/www/html/index.html
root@207950dd30a0:/#
```



Hola Mundo desde Nginx en Docker

Servidor funcionando correctamente